

- 1 (a) Table 1.1 shows three of the processes by which substances in solution can move across cell membranes. It also lists five statements that may apply to each of these three processes.

Complete Table 1.1 to show which of the statements apply to each of the three processes shown.

Use a tick (✓) to show that the statement applies or a cross (X) to show that the statement does not apply.

Each box must contain a tick or a cross.

The first row has been completed for you.

Table 1.1

statement	process		
	active transport	facilitated diffusion	simple diffusion
movement of oxygen into a red blood cell	X	X	✓
occurs in both animal and plant cells			
uses carrier proteins			
movement of non-polar molecules between the fatty acid tails of the phospholipid molecules			
movement of ions down a concentration gradient			

[4]

(b) Fig. 1.1 is a simplified diagram representing a transverse section of part of a young root. The diagram is not to scale.

(i) On Fig. 1.1 draw a label line and label with the letter **C** to identify the Casparian strip. [1]

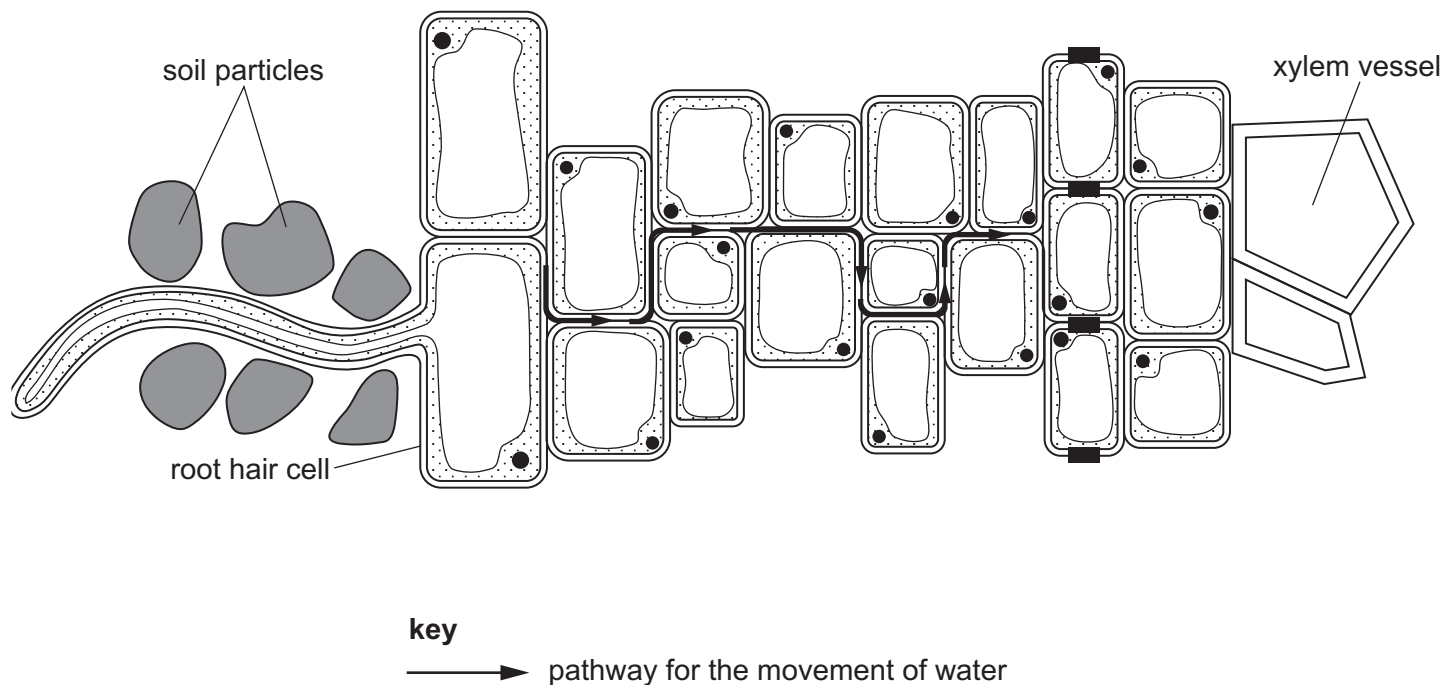


Fig. 1.1

(ii) Root hairs measure approximately $5\mu\text{m}$ in diameter and $500\mu\text{m}$ in length.

Explain how this adapts root hairs for the absorption of water.

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 [1]

(iii) Name the pathway for the movement of water shown by the arrows in Fig. 1.1.

..... [1]

(c) Water enters the xylem vessels shown in Fig. 1.1.

Explain how water moves up the xylem vessels to the leaves in a continuous column.

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..... [3]

[Total: 10]